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A Monolayer Carbon Nitride on Au(111) with a High Density of Single Co Sites

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ABSTRACT Carbon nitrides that expose atomically dispersed single-atom metals in the form of M-N-C (M = metal) sites are attractive earth-abundant catalyst materials that have been demonstrated in electrocatalytic conversion reactions. The catalytic performance is determined by the abundance of N-doped sites and the type of metal coordination to N, but challenges remain to synthesize pristine carbon nitrides with a high concentration of the most active sites and prepare homogeneously doped materials that allow for in-depth characterization of the M-N-C sites and quantitative evaluation of their catalytic performance. Herein, we have synthesized and characterized a well-defined monolayer carbon nitride phase on a Au(111) surface which exposes an exceedingly high concentration of Co-N₄ sites. The crystalline monolayer carbon nitride, whose formation is controlled by an on-surface reaction between Co atoms and melamine on Au(111), is characterized by a dense array of four- and six-fold N-terminated pockets, whereof only the four-fold pocket is found to be holding Co atoms. Through detailed characterization using scanning tunneling microscopy, X-ray photoelectron spectroscopy, and density functional theory modeling, we determine the atomic structure and chemical state of the carbon nitride network. Furthermore, we show that the monolayer carbon nitride structure is stable and reactive towards the electrocatalytic oxygen reduction reaction in alkaline electrolyte, with a quantitative performance metric that significantly exceeds comparable M-N-C-based catalyst types. The work demonstrates that high-density active catalytic sites can be created using common precursor materials, and the formed networks themselves offer an excellent platform for onward studies addressing the characteristics of M-N-C sites.

Main Text

Heteroatom-doped carbon-based materials constitute a promising group of new sustainable compounds for applications in green energy conversion by catalysis thanks to their favorable abundance, electrical properties, and high chemical versatility. Functionalization of carbon-based materials by nitrogen has attracted particular attention for the synthesis of “M-N-C” sites which are single atom transition metal (M) centers coordinated to N within a carbon matrix, shown to be active sites for electrocatalytic water splitting, CO₂ conversion, and hydrogen peroxide formation^{1–3}. Although the exact M-N-C site structure, location, and density are predicted to be critical to the catalytic performance⁴, the primary approaches taken during the synthesis of N-doped carbon materials typically generate a complex material with a multitude of different sites and a low active site density⁵. Single metal coordination to N in these materials can, in principle, be realized with significant variation in structure and coordination^{6–11}, but theoretical studies have pointed out that M-N-C sites with four-fold nitrogen coordination to metals such as Co, Fe, Ni, and Mn (M-N₄) are more stable and active than other types of M-N coordination^{12–16}.

Strategies to synthesize N-doped carbons with a uniform distribution of well-defined N-pockets available for metal coordination and improved site density include both bottom-up synthesis pathways^{17–19}, improved N-doping procedures¹³ or the use of metal-organic-framework (MOF)-like materials with a well-defined and high concentration of coordination sites²⁰. However, these, in general, suffer from low nitrogen concentrations, limiting the availability of M-N₄ coordination sites. Another potentially attractive route for M-N-C site formation involves “graphitic” carbon nitrides (g-C₃N₄), which have already been widely demonstrated as photocatalysts^{21–23} or in electrocatalysis^{24,25}. The use of g-C₃N₄ as N-doped graphene may be attractive due to easy synthesis and high C-N content. Yet, a general method of exploiting the potentially large

availability of N-containing coordination pockets for M-N-C type catalyst with a high active site density remains to be demonstrated. Two-dimensional sheets of $g\text{-C}_3\text{N}_4$ are formed by the polymerization of precursors such as melamine ($\text{C}_3\text{H}_6\text{N}_6$). Upon heating up to 900 K, the precursor will condense while releasing ammonia. Here, various intermediate phases form through the joining of new sp^2 C-N bonds. At first, melam units and then melem units are formed. Then as temperature increases further, the melem units polymerize to yield melon chains, which are eventually linked together, and finally, $g\text{-C}_3\text{N}_4$ -sheets are obtained^{26,27}. The structural diversity induced by such synthesis routes means that $g\text{-C}_3\text{N}_4$ may expose a significant structural variation in the type of N-pockets that could potentially coordinate catalytically active metals. Importantly, however, none of the standard $g\text{-C}_3\text{N}_4$ polymerization routes, leading to triazine-bonded networks, generate pockets with a structure that contains the predicted optimum M-N-C-type sites, with four-fold coordination to metals. Thus, obtaining a high concentration of M-N₄ starting either from carbon- or nitrogen-rich precursors has proven difficult.

In this work, we react metallic Co and melamine as precursors directly on a Au(111) surface and demonstrate the synthesis of a monolayer carbon nitride structure which directly favors the formation of a high density of single Co atom sites with 4N-coordination. Unlike $g\text{-C}_3\text{N}_4$, the carbon nitride network has a C_9N_{10} stoichiometry and is joined from melamine units by both sp^2 -bonded C-C and C-N groups which generate a structure with both four-fold and six-fold coordination possibilities for Co to lattice N within the carbon-nitride sheet. Using scanning tunneling microscopy (STM), and X-ray photoelectron spectroscopy (XPS), combined with density functional theory (DFT), we propose a model for the formed carbon nitride and show that Co is preferentially located within pockets with four-fold nitrogen coordination. We furthermore characterize the electrocatalytic properties of the carbon nitride monolayer in the oxygen reduction

reaction (ORR) and find that the structure is stable over several cycles with an activity reflecting the high abundance of active Co sites. Our work thus demonstrates that monolayer carbon nitrides with an extraordinarily high concentration of Co monomers active in the ORR reaction can be synthesized.

Results and Discussion

Formation of the Cobalt Carbon Nitride

Figure 1a displays a large-scale STM image of the monolayer carbon nitride formed by reacting Co metal from a physical vapor deposition (PVD) source with melamine on the terraces of a pristine Au(111) surface. The monolayer structure is resolved by STM in the insert and reflects a dense and highly ordered network with hexagonal symmetry. The periodicity of the hexagonal lattice is $11.5 \text{ \AA} \pm 0.5 \text{ \AA}$, and the layer has an apparent height of $1.3 \text{ \AA} \pm 0.4 \text{ \AA}$ relative to the gold, consistent with a monolayer structure.

Various synthesis approaches were investigated. The monolayer formation was observed to be sensitive to deposition rates, substrate temperature, and the sequence in which Co and melamine are deposited. The most efficient method leading to a uniform and crystalline monolayer carbon nitride involved slow co-deposition of cobalt (0.01 monolayer(ML)/min) together with melamine in high excess relative to Co onto the clean (reconstructed) Au(111) surface held at 450 K. Direct deposition followed by high-temperature annealing of melamine on Au(111) did not result in carbon nitride. Melamine forms a rather relatively well-ordered and dense molecular ad-layer on Au(111)^{28–30} (Figure S1), but we found that the temperatures necessary for the activation of the polymerization reaction (900 K) are much higher than the desorption temperature of melamine

(375 K, Figure S2). In contrast, the fact that a carbon nitride readily forms by co-deposition of Co at a temperature of 450 K, much lower than the normal synthesis temperature of carbon nitride, indicates that the co-deposited Co directly participates in the formation of the monolayers. Furthermore, a mix of Co nanoparticles on Au(111) and melamine is not reactive toward the formation of any monolayer structures (Figure S3). Importantly, this shows that the supply of reactive single cobalt atoms (Co_1) species during PVD is essential for the synthesis and that Co_1 species are likely dispersed and bonded directly into the observed carbon nitride structure³¹. We, therefore, refer to the carbon nitride monolayer as ‘CoCN’.

To determine the chemical composition and bonding of the CoCN monolayer we performed a series of synchrotron XPS measurements focusing on N 1s and C 1s core level shifts of melamine (Figure 2a,b) and CoCN (Figure 2c,d). Figure 2a,c first compares the pristine N 1s spectrum for melamine on Au(111) with the corresponding N 1s spectrum of CoCN formed by the reaction of melamine with Co_1 on Au(111). The melamine molecule contains two types of nitrogen (see ball model in Figure 2a), an aminic (C-NH_2) (blue dashed curve) and a pyridinic nitrogen (C-NH-C) (orange dashed curve). The atomic ratio between the two species is 1:1, however, the spectrum does not fully reflect this, as the maximum intensity of the two peaks appear slightly different. This is explained by melamine adsorbing in a bi-layer structure when the coverage exceeds a monolayer, with energies for the second layer melamine shifted 0.5 eV to higher binding energy³⁰. The fitted binding energies are 397.8 eV for the first layer pyridinic component and 398.8 eV for the aminic component, respectively.

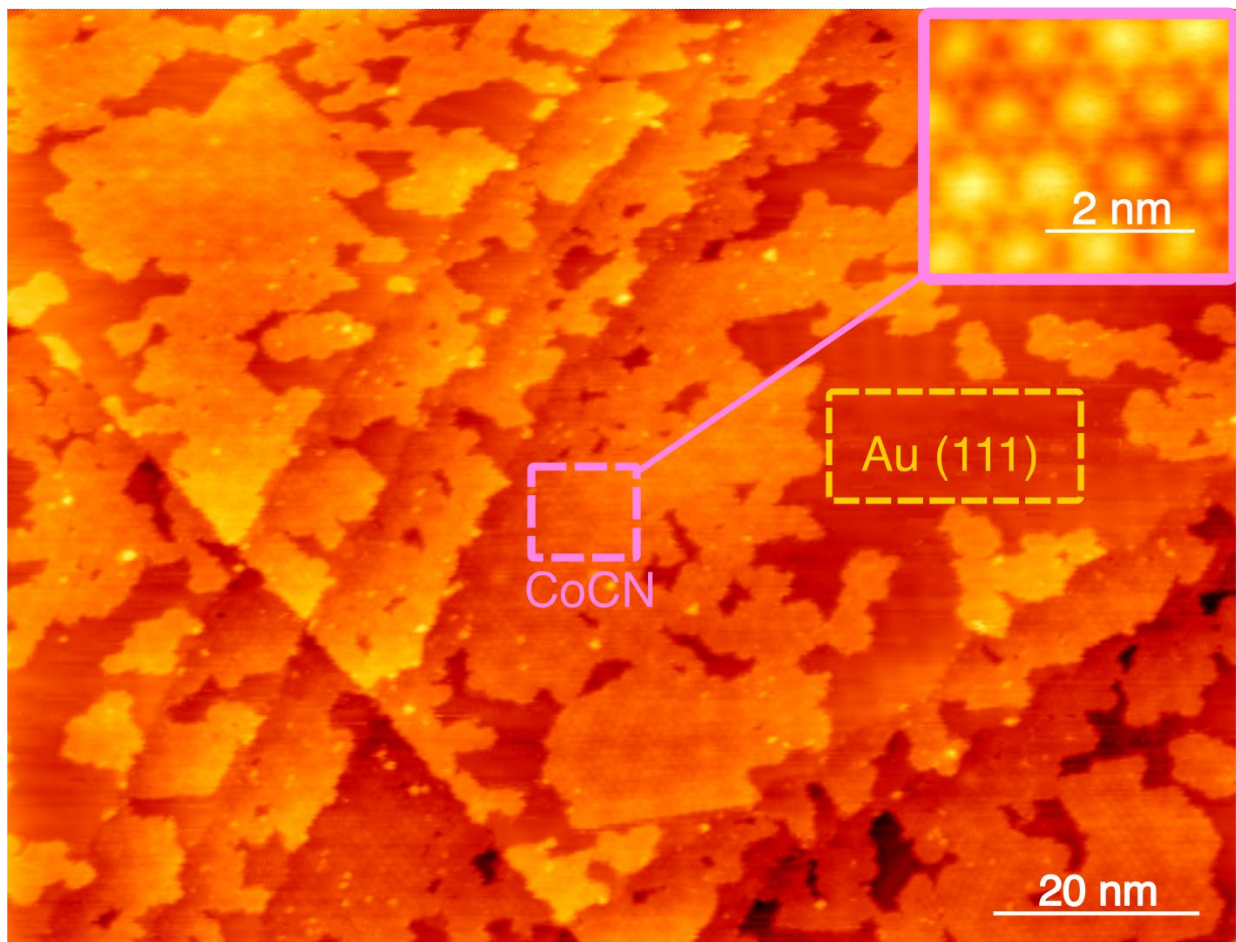


Figure 1: Cobalt carbon nitride layer on Au(111). Large-scale STM image of 0.7 ML cobalt carbon nitride formed by co-deposition of Co metal and melamine on Au(111). The dashed squares indicate regions containing the cobalt carbon nitride monolayer (purple) and bare Au(111) (yellow). The inset is a high-resolution magnified view, showing the periodicity and symmetry of the cobalt carbon nitride structure. STM parameters: $I_t = 0.46$ nA, $V_t = 154$ mV (large-scale), and $I_t = 0.30$ nA, $V_t = 114$ mV (inset).

A significant change in the N 1s spectrum (Figure 2c) was observed after CoCN formation. Instead of the two-peak structure of pure melamine, the N 1s peak structure now reflects four different N species, each with a distinct chemical environment, of which only the position of the aminic peak

in melamine (blue dashed curve) is conserved. Together with a dominant center peak (magenta dashed curve), which is slightly shifted from the main pyridinic peak in melamine, a new pronounced peak appears on the low binding energy side (purple dashed curve) and a shoulder appears on the high binding energy side (green dashed curve). The substantial decrease in the intensity of the aminic peak coupled with the chemical shifts of the N species are a strong indication that a network has been formed by the reaction of the terminal aminic groups in melamine. A direct comparison of Figure 2c with the corresponding N 1s spectra recorded for *g*-C₃N₄ powder³² reveals that the N 1s spectrum in CoCN is not compatible with the spectrum from the *g*-C₃N₄ powders since these feature no species with a lower binding energy than the pyridinic species, while the CoCN spectrum contains a significant peak at 396.9 eV. Based on reference experiments (Figure S4), it can be excluded that the low energy peak originates from an interaction between melamine monomers and cobalt nanoparticles. Therefore, we conclude that the synthesis pathway of CoCN is significantly different from the conventional model leading to the typical melem-based carbon nitrides.

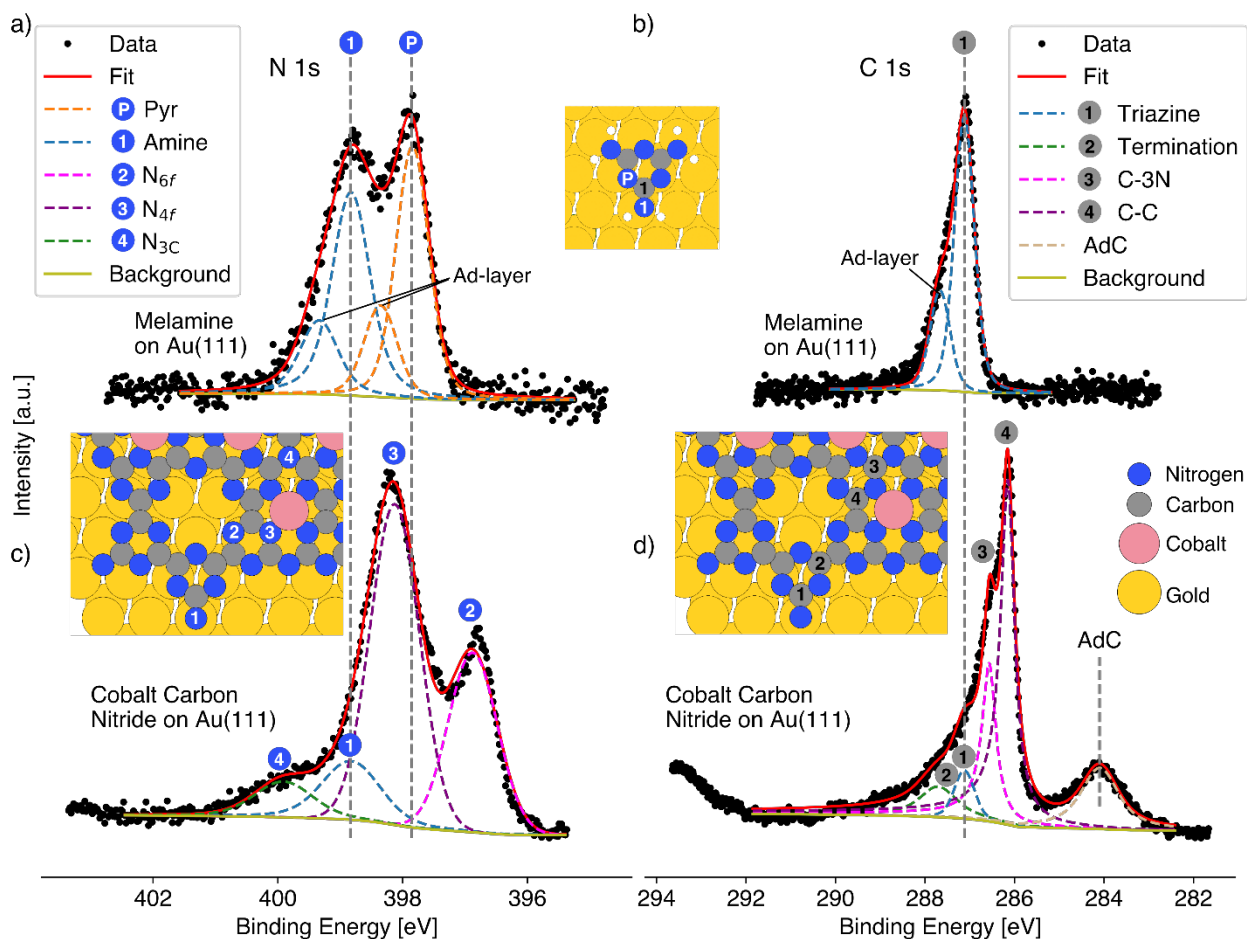


Figure 2. High-resolution XPS data. XP spectra recorded at the FlexPES beamline of MAX-IV comparing the melamine precursor (top a,b) to the cobalt carbon nitride (bottom c,d). (a,b) The N 1s and C 1s for 1.3 ML melamine on Au(111). (c,d) The N 1s and C 1s for ~0.5 ML CoCN on Au(111). (a,c) are measured at a photon energy of 500 eV, while (b,d) are measured at a photon energy of 386 eV. Vertical grey lines are added to guide the eye at peak positions found in the fitting of melamine spectra. The insets sketch the corresponding structures with signal assignment to individual species. In (d), the peak indicated at 284.2 eV is assigned to adventitious (AdC) carbon which is not part of the CoCN structure.

The difference to $g\text{-C}_3\text{N}_4$, which has just one C 1s peak³² in its spectrum, becomes further evident from inspection of the C 1s spectra in Figure 2b,d which again compares the melamine on Au(111) and the CoCN monolayer. For melamine on Au(111) (Figure 2b), the C 1s signal comprises a main peak at 287.1 eV originating from sp^2 C bonded to N and a smaller peak at 0.5 eV higher binding energies from second layer melamine³⁰. After the formation of the cobalt carbon nitride, there is a significant broadening and shift of the C 1s structure. Peak fitting shows that two new peaks appear on the low binding energy side at binding energies that are usually associated with C-C bonds, whereas only a minor peak is present for the original triazine bond. This lowering of the binding energy surprisingly indicates the formation of carbon-carbon bonds. Furthermore, the peaks are slightly asymmetric. The asymmetry of the peaks is also commonly seen for highly oriented pyrolytic graphite³³ and other functionalized graphitic carbons³⁴, which supports the assignment of a C-C bond. Again, the C 1s peak structure of the CoCN monolayer differs from $g\text{-C}_3\text{N}_4$, where only one carbon species is present³². A measurement of the Co 2p region reveals that cobalt is present as both Co^0 and Co^{2+} on the surface (Figure S5). Based on reference measurements we assign the Co^{2+} to cobalt in the CoCN structure, and Co^0 to residual Co nanoparticles on the surface formed from unreacted Co that has nucleated on the surface (see also Figure S6).

Structural Model for the Cobalt Carbon Nitride

High-resolution STM images of the CoCN monolayer (Figure 3a), reveal it as a hexagonal grid composed of hexagonal building units (white outline), each imaged with six protrusions at the corners of the hexagons and with the center of the hexagon being slightly depressed. Further, depressions are seen along the packing direction of the grid at the lateral interface between each hexagon unit. From its symmetry, this structure of the CoCN is not in accordance with any

proposed g-C₃N₄ monolayer structure^{21,26,27}, since these feature triangular subunits in the rhombic unit cell, and a minimum repeat distance of 7.1 Å^{35,36}, incompatible with the symmetry and periodicity (11.5Å) of the observed CoCN monolayer.

We therefore propose a different model for a cobalt carbon nitride monolayer, that explains the STM and XPS observations (Figure 3b–c). Figure 3b shows just the carbon-nitrogen backbone, while Figure 3d also includes the position of Co atoms. A chemical sketch without the substrate is furthermore illustrated in Figure S7. The proposed model has an overall Co₃C₁₈N₂₀ stoichiometry. In Figure 3b, the carbon-nitrogen backbone is formed by joining together 6 triazine rings (C₃N₃) (green hexagon) with C-C bond formation into a “super-ring” structure (white hexagon) that forms the hexagonal building block of the monolayer structure, consistent with the structure observed in STM. These super-rings are then joined together through bridging nitrogen atoms (corners of white hexagons), giving the structure a nominal periodicity of 11.9 Å, in agreement with the experimental observation. The CoCN is nearly commensurate with the underlying Au(111) structure, with a possibility for the (4×4) periodicity as indicated by the model (back rhomb). A (4×4) periodicity would imply a length mismatch of 3.4%. The epitaxy is reflected in an apparent preferred orientation of the CoCN monolayer along either of the threefold high symmetry directions of the Au(111) surface (Figure S8).

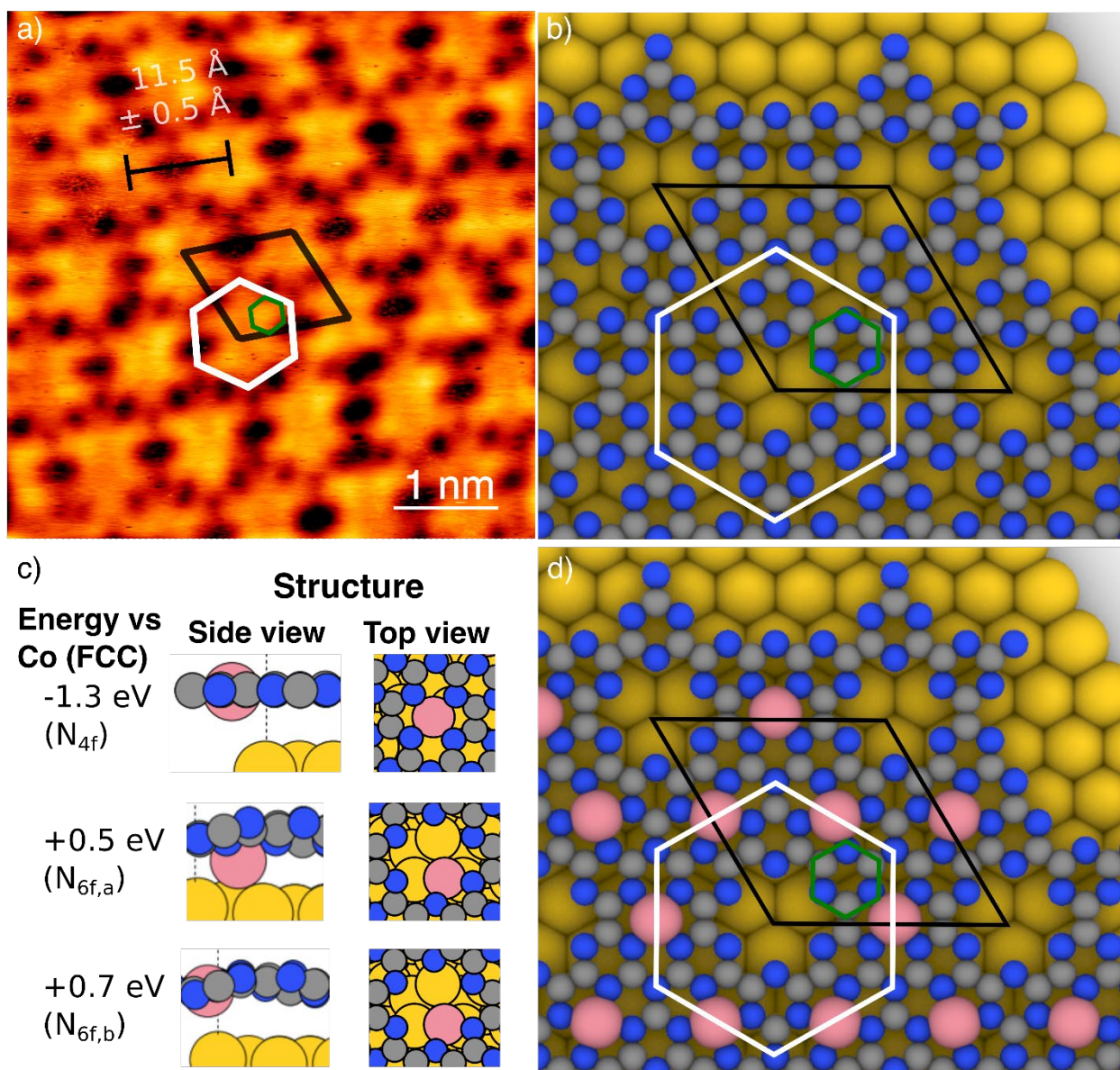


Figure 3. High-resolution STM image of the CoCN on Au(111) and atomistic modeling. (a) STM image acquired at $I_t = 0.390 \text{ nA}$ and $V_t = 50.4 \text{ mV}$, displaying an imaging mode with hexagonal CoCN units. (b) The atomistic modeling of carbon nitride backbone, shown is carbon (grey), nitrogen (blue), and the gold substrate (gold). (c) Chemical structure and adsorption energy of CoCN on gold substrate. (d) The atomistic modeling of CoCN with cobalt (pink) added in the four-fold pockets.

The CoCN monolayer exposes two types of N-terminated pockets, a six-fold pocket in the center of the super-ring and a four-fold pocket between the super-rings. Following the structural model, the protrusions at the corner of each of the hexagons in the STM image are the triazine rings, while the slight depression at the center is the central six-fold pocket, and the holes along the packing direction are the four-fold pockets. To examine the preferential site of Co₁ in the structure, we performed DFT calculations (Figure 3c). The calculations are performed for a monolayer of CoCN on top of Au(111) (See Methods). Here the adsorption energy (ΔE_{Co}) of Co₁ in different configurations is compared to that of cobalt in the FCC structure, with a negative value corresponding to favorable adsorption. Co insertion into the four-fold pocket results in a stable Co-N₄ configuration, with cobalt atoms incorporated here being favorable at $\Delta E_{Co} = -1.3$ eV. For cobalt insertion with six-fold coordination, two configurations are possible, differing by the location of the Co relative to the Au(111) (Figure 3c). In both cases, Co insertion occurs adjacent to 2 of the N on the side of the six-fold pocket, which is unstable ($\Delta E_{Co} > 0$). Furthermore, the cobalt can either be displaced towards a gold hollow site (not in the CoCN plane), being $\Delta E_{Co} = +0.5$ eV (denoted N_{6f,a}). Alternatively, the cobalt can coordinate in-plane with the CN lattice, being slightly less favorable at $\Delta E_{Co} = +0.7$ eV (denoted N_{6f,b}). Clearly, only a filling of the four-fold pocket is favorable, whereas no Co is expected in the six-fold pocket. This is consistent with the observation of a central depression within the hexagon in Figure 3a, but we note that in other STM images (for example Figure 1a (inset) and Figure S9), the apparent height of the center was sensitive to the STM tip state being imaged either as a hole or protrusion, which indicates that the electronic structure of the conjugated network is important for the overall STM contrast.

Additional confirmation for cobalt inclusion in the four-fold pocket is found from the analysis of the chemical shift in the XPS data shown in Figure 2c. The N 1s spectrum is described by four species (Table S1). We assign the 399.9 eV feature to the bridging N_{3C} nitrogens (green dashed line), and the 398.8 eV to amines (blue dashed line) as in the melamine spectrum. The pure pyridinic peak is no longer present in CoCN. However, the shift between the pyridinic energy and the main peak (N_{4f}, purple dashed line) seen in CoCN at 398.1 eV is 0.3 eV to higher binding energies. We assign this shift to the filling of the four-fold pocket with Co. Porphyrins, which expose structurally identical 4N pockets, have been prepared and characterized both with (i.e., Co-N₄) and without metals. In literature³⁷, a shift of 0.5 eV is found between the N 1s species of an empty and a Co-filled 4N-pocket in powders. We conducted reference experiments depositing porphyrins on Au(111). In agreement, these show a shift of 0.3 eV towards higher binding energy in the N 1s spectrum as the empty N₄ pocket is filled (see Figure S10), matching the change seen for CoCN.

Peaks with energies lower than the pyridinic peak are uncommon in carbon nitride systems³². Therefore, DFT calculations were performed to examine the XPS binding energies of species next to a cobalt pocket filled to varying degrees (Table S2). Contrary to a simple electronegativity-driven trend in chemical shift, we find that cobalt incorporation leads to a higher binding energy of the surrounding nitrogen atoms. This agrees with our XPS measurements and supports the assignment of N_{4f} species coordinating to a cobalt atom. The predicted energy difference between the N 1s signal of a N_{4f} coordinated to cobalt and a N_{6f} in an empty pocket is 1.3 eV, closely matching the observed difference of 1.2 eV in Figure 2c. Thus, we conclude that the six-fold pocket is empty, in agreement with the adsorption energies in Figure 3c, and that the peak at 396.9 eV originates from nitrogen species in the six-fold pocket.

The proposed model also matches with the carbon XPS data, which is likewise described by four different species (Table S3). The largest peak is C-C bonded carbon (purple dashed line), and then going towards increasing binding energy, we assign the peaks to C-3N (pink dashed line), triazine-carbon (blue dashed line), and finally termination states (green dashed line), reflecting that carbon atoms at the edge of the system should not exhibit energies that perfectly resemble any of the other proposed states.

For N 1s, we quantitatively expect the sheet to have a 12:6:2 ratio between the nitrogens in the four-fold pockets (N_{4f}), six-fold pocket (N_{6f}), and N_{3C} . This is in excellent correspondence with observed ratios of 11.7:6.6:1.7, obtained from peak-fitted areas in Figure 2c. Similarly, the C-C and C-3N species in the C 1s spectrum are expected to have a 12:6 ratio, and experimentally we find 12.4:5.6, again in excellent agreement. The remaining two species only exist on the edge of the structure, and thus, their ratios are related to the extent of layer formation. We note that we have not considered models formed through an opening of the triazine ring in detail. Triazine rings are generally regarded as sturdy, owing to the aromatic C=N bonds³⁸. Furthermore, the protruding hexagon corners in Figure 3a, together with the position of the N_{4f} peak in the N 1s spectrum, suggest that the rings are intact. Thus, we arrive at the model shown in Figure 3c, the same as in Figure 3b, but now with the four-fold pockets filled with cobalt atoms. The exact position of the Co in the pocket and the bond lengths of the CoCN structural model could be further elucidated using either quantitative low energy electron diffraction (LEED)³⁹, photoelectron diffraction or high-resolution non-contact atomic force microscopy (NC-AFM)⁴⁰. The formed CoCN has a high loading of Co in Co- N_4 sites at 7.3 at% (26 wt%). Importantly, this represents an exceedingly high metal (Co) site density compared with reported single atom catalysts^{5,41}, which typically exhibit a concentration below 5 wt% or 1 at%.

Electrochemical Characterization

To investigate the electrocatalytic properties of the CoCN network towards the ORR, we performed cyclic voltammetry measurements using the CoCN sample as a working electrode. To perform these measurements, a pre-characterized CoCN sample was extracted from vacuum and installed in a dedicated three-electrode setup (see Methods). After recording of cyclic voltammograms, the sample was returned to vacuum for inspection. The voltammetric results indicate that the CoCN monolayer is stable and performs better quantitatively than comparable M-N-C catalysts due to the combination of high turnover frequency (TOF) and a high concentration of Co sites. To see this, Figure 4a shows the electrochemical data obtained from a 0.7 ML CoCN on Au(111) (as confirmed with STM, Figure S11) compared with a clean Au(111) surface in an oxygen-saturated 0.1 M NaOH solution. The measurements were conducted in oxygen-saturated media. The data shows an onset potential of 0.83 V vs RHE (defined as the potential to reach $-0.025 \text{ mA cm}^{-2}$ in current density) for ORR. The current density reaches a maximal magnitude of -1.86 mA cm^{-2} at 0.55 V vs RHE before oxygen depletion at the electrode surface sets in. Figure S12 shows that the performance of CoCN stays stable over five cycles. Furthermore, the presence of N 1s, Co 2p, and C 1s signals after electrochemistry also confirms the persistence of the CoCN on the Au(111) (Figure S13). We note that the CoCN exhibits an oxidation peak at 1.1 V vs RHE, which corresponds to the oxidation of the residual Co NPs on the surface. However, this does not impact the catalytic performance as changes in the state of the Co NPs should lead to changes in the catalytic performance if they were the active units. To confirm that the activity does not originate from the Co NPs, Figure S14 shows that catalytic activity across 3 samples scales with

N content (i.e. the amount of CoCN) rather than Co content. Furthermore, a separate reference measurement starting with metallic Co NPs on Au(111) (Figure S15) shows the CoCN is significantly more active and stable than metallic Co. Therefore, the ORR activity is assigned to the Co-N₄ catalytic unit.

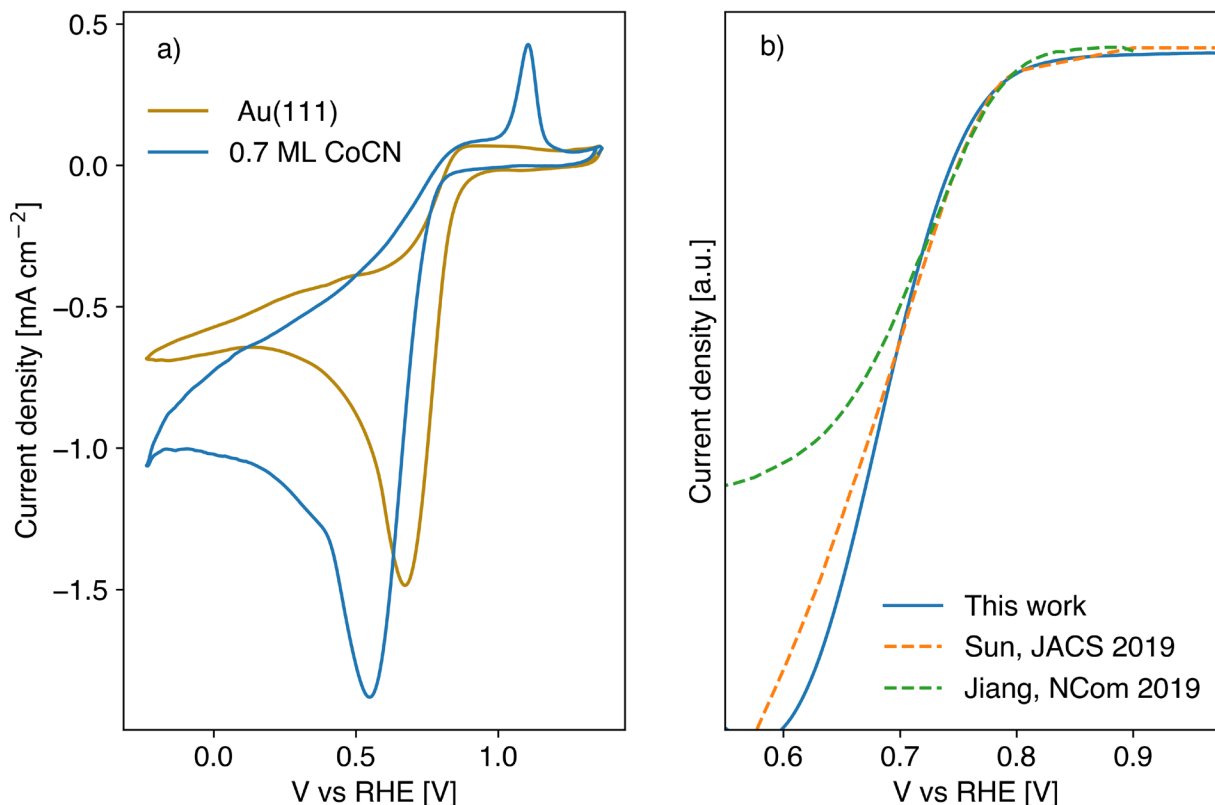


Fig. 4. Electrochemical results for the ORR. (a) Comparison of cyclic voltammograms of CoCN (blue) and the Au(111) (gold) substrate in oxygen saturated 0.1 M NaOH with a sweep rate of 100 mV s⁻¹. (b) The region of a cyclic voltammetry scan showing the regions of onset and exponential increase. The data is compared with two other studies^{42,43} on Co-N₄ active sites scaled to show the similarity of onset. Data adapted with permission from reference ⁴² Copyright 2019 American Chemical Society.

To investigate if part of the catalytic activity originates from the Au(111) substrate, a reference measurement of Au(111) was carried out. Figure 4a shows that the Au(111) has an earlier on-set potential and comparable maximal magnitude of the current density. However, the Au(111) substrate is not responsible for the observed current in CoCN since the remaining 30% uncovered Au(111) should then result in current densities of -0.05 mA cm^{-2} at 0.86 V vs RHE; there is no discernable current at this potential. The absence of Au(111) activity for the CoCN signal is explained by the CoCN growing preferentially at step edges, which appear to be the active sites on the Au(111) substrate⁴⁴.

Importantly, ORR activity studies of Co-N₄ sites created in carbon nanotubes⁴³ or MOFs⁴² show H₂O₂ production at onset potentials of 0.82 and 0.83 V vs RHE, respectively, achieving high selectivity and current. These onset potentials are practically identical to the one measured in this study, i.e., 0.83 V vs RHE (see Figure 4b). Moreover, the development of the current is alike as the potential is scanned to more reductive potentials. The difference in maximal current reached between this study and previous ones is attributed to differences in loading and mass transport. Because of the similarity of the active sites in this and previous studies^{42,43}, we assign the ORR activity of the CoCN/Au(111) in this work to H₂O₂ production via the 2e⁻ pathway.

With an accurate atomic scale model for the CoCN, the number of active Co-N₄ sites per unit area can be estimated with unprecedented accuracy in our case. A turn-over frequency (TOF) can be determined to be $1.50 \text{ e}^- \text{ s}^{-1}$ at 0.8 V vs RHE (before mass transport limitations set in) while exceeding $40 \text{ e}^- \text{ s}^{-1}$ at 0.55 V vs RHE. Both are significantly higher than TOFs commonly reported for similar M-N-C materials⁴⁵⁻⁴⁷. These numbers thus reflect the intrinsic activity of the CoCN as a planar model catalyst, but we note that compared with other types of M-N-C catalysts additional effects may contribute, such as enhanced electrical conductivity through the Au(111) to the

catalytic layer and a reduced effect of mass transport at a planar surface, since no pores have to be traversed. The electrochemical measurements confirm a high activity and stability of the synthesized CoCN.

Discussion

The cobalt carbon nitride monolayers formed on a single crystal substrate are readily investigated using STM and XPS, but we expect that the planar and well-defined nature of the monolayers demonstrated here and will enable the use of a wider range of surface science technique techniques, including *operando* spectroscopy and microscopy, to elucidate the fundamental catalytic nature of M-N-C type active sites. Our preliminary work shows that structurally identical NiCN and FeCN monolayers can be created, which we will report in a forthcoming study. Similar reactions have also been performed by reacting melamine with a Cu (111) substrate⁴⁸. This type of synthesis was shown to form a catalytic Cu-N₂ unit⁴⁹.

Importantly, our results also demonstrate that well-defined monolayer carbon nitride phases that differ significantly from *g*-C₃N₄ in their structure and capability to coordinate Co species, can be realized. Notably, the cobalt carbon nitride is formed already at 450 K, which is far lower than that used in typical pyrolysis methods for *g*-C₃N₄. The prospects of generalizing this approach to transition metal-doped carbon nitride synthesis as a bulk material, relies on control over synthesis conditions. Importantly, the Co₁ appears both as a catalyst in the formation of the monolayer and as the active species in the formed layer in the ORR reaction. The role of the Co₁ atomic species is apparent when considering experiments with sequential dosing. If cobalt is added first, forming nanoparticles, followed by melamine, no formation of the CoCN layer takes place, while if melamine is added first, followed by cobalt, there is some layer growth (Figure S3). Thus, methods

for bulk synthesis would rely on agents that contain atomically dispersed Co₁ species as a chemical precursor. The Au(111) seems to have a facilitating role for the synthesis of the monolayers since bilayer formation was not observed in our experiments. However, the conjugated nature of the C-N and C-C bonds, suggests that the CoCN monolayers can form without the Au(111) substrate. During synthesis the Au(111) is therefore considered as an interface that increase the chance that Co and melamine can react already at a low temperature of 450 K. Once a full monolayer is formed, additional supply of Co atoms will not lead to reaction with melamine, but instead the formation and nucleation of Co nanoparticles. The presence of two peaks in the Co 2p XP spectra (Figure S5) supports this interpretation, i.e., the Co⁰ originates from excess Co nucleated as cobalt nanoparticles (NPs), while the Co²⁺ originates from cobalt incorporated into the CoCN. Future work, for example for synthesis on graphite (HOPG) or another monolayer phase such as graphene/Ir or Ru, could help elucidate the role of other conducting substrates. Moreover, we anticipate that the atomically dispersed Co atoms in carbon nitride monolayers may have interesting electronic or magnetic properties, that await investigation in future work.

Conclusion

Based on a combined experimental and theoretical approach, we demonstrate the possibility of forming a CoCN catalyst with a high abundance of Co-N₄ active sites. An atomistic model was provided for the CoCN from STM, XPS, and DFT calculations. The CoCN arises through the formation of an extended network based on triazine units and forms two types of pockets; a four-fold pocket is filled with cobalt, and a six-fold pocket which remains empty. The formation of the layer is mediated by the reactive Co₁ species deposited and the Au(111) surface. The formed CoCN is stable in alkaline electrolyte and shows catalytic reactivity towards the ORR and reaches a TOF of 1.5 s⁻¹ at 0.8 V vs RHE. The high TOF combined with the high density of atomic cobalt sites

allows for excellent activity. Our results offer an experimental framework for synthesizing and investigating the fundamental properties of M-N-C materials containing M-N₄ active sites.

Materials and Methods

Laboratory STM and XPS Experiments: Our STM and laboratory-source XPS measurements have been conducted in an ultra-high vacuum system (UHV) with a base pressure better than $1 \cdot 10^{-10}$ mbar. The STM is an Aarhus-type STM (Aarhus 150 SPECS GmbH) equipped with a tungsten tip. The lab source XPS (SPECS XR 50) is a dual anode system with Al K α and Mg K α radiation. All lab source spectra reported here were collected using the Al K α anode. The system is equipped with a SPECS Phoibos 100 detector. The XPS energies were calibrated by setting the Au 4f $_{7/2}$ binding energy position to 84.0 eV. All STM and XPS measurements were performed at room temperature.

Synchrotron XPS experiments: Synchrotron XPS experiments were performed at MAXIV at the FlexPES beamline. Data was collected with a slit of 25 μ m and a pass energy of 20 eV. All spectra were collected at two excitation energies; 500 eV and 700 eV for N 1s, while 386 eV and 586 eV for C 1s. Energies were calibrated by measuring the Fermi Edge (–1 to 1 eV) immediately after the core level spectra. The core level spectra were then shifted to according to the energy difference between the center of a sigmoid fitted to the Fermi Edge signal and 0 eV.

XPS Analysis: XPS analysis was performed in python using the lmfit⁵⁰ package. Data was fitted with a combination of Voigt functions as implemented in Scientific Python and the Doniach Sunjic lineshape as implemented in HyperSpy. Static backgrounds were employed using Shirley backgrounds all spectra. All four spectra were fitted as a single fit, where data was normalized, such that the integrated intensity for each spectrum was equal. During this quadruple fit certain parameters were fixed to each other.

Within each spectrum, the gaussian full width at half maximum (FWHM) was fixed to be the same for all species, while the Lorentzian FWHM was allowed to vary. The alpha-parameter for

the Doniach-Sunjic lineshape was fixed to be the same for both the C-C and C-3N peak. The melamine add-layer coverage was set to be the same in both the C 1s and N 1s fit. The positions of all species were held constant across the fits (i.e. the amine in melamine must have the same binding energy as the amine species in CoCN). The adlayer in melamine was fixed to have the same FWHM as the baselayer.

Sample preparation and synthesis: The sample used was a Au(111) single crystal from Mateck, with a diameter of 7 millimeters. Before synthesis and characterization, the sample was cleaned by successive cycles of sputtering and annealing. Sputtering was performed by Ar ions with an accelerating voltage of 1.5 kV, typically for 30 minutes per cycle. Annealing was done at 830 K typically for 15 minutes per cycle. Starting from a clean surface (confirmed through STM and XPS) the samples were prepared. Melamine was evaporated using a homebuilt molecular evaporator, consisting of a melamine filled cylindrical glass crucible with a filament wrapped around. The temperature was measured using a K-type thermocouple at the bottom of the crucible. Deposition of melamine was done at crucible temperatures of 380 K. Cobalt was evaporated using a MANTIS E-BEAM evaporator (QUAD-EV-C) with four pockets. To obtain the CoCN layer melamine and cobalt was co-deposited onto the clean Au(111) surface, keeping the surface at an elevated temperature of 450 K.

DFT Calculations: The atomic structures are created in the ASE software⁵¹. The gold (Au) structures is built with a $4\times 4\times 3$ unit cell with the Au(111) facet exposed and with the two lower layers fixed to mimic bulk Au. The CoCN structures are built by taking 6 melamine rings as starting point and removing parts of these. Finally, a vacuum of min 20 ang is set. For relaxation computations we use the GPAW code^{52,53} with the following settings: (3, 3, 1) k-point, a grid spacing of 0.18, with the BEEF-vdW functional⁵⁴ and relax the structures to below 0.05 eV/atom.

After relaxating the CoCN structure, the stability of the Cobalt in the structure is calculated from the following equation:

$$\Delta E = E_{structure}^{Co} - E_{Co(s)} - E_{structure}$$

Using CoCN structures the changes in the XPS nitrogen edge (Table S2) is calculated different atoms in the structure using DFT Delta Kohn-Sham calculation. Here the total energy difference between the ground state and the first core excited state is calculated by generating a full core hole PAW setup and using the equation:

$$E_{XPS}^{N\ 1s} = E_{DFT}(ground\ state) - E_{DFT}(N\ 1s\ core\ hole)$$

This method gives the energy of generating the core hole in a specified atom. While the absolute energy can be off by eVs, the relative energies between different N species in the structure are more accurate, which is why in Table S2 we use one value as reference with experiment.

Electrochemical measurements: Electrochemical measurements were carried out in a home-made H-type cell using an Metrohm Autolab potentiostat/galvanostat (PGSTAT302). Inside the cell, it was equipped with an anion-exchange membrane (Selemion AMV, AGC) to sperate the anodic and cathodic compartments. The sample acted as the working electrode, with a working area of 28.3 mm² while a platinum foil in the anodic side served as the counter electrode and a commercial leak-free Ag/AgCl reference electrode (LF-1.6-100, Innovative Instruments Ltd.) was placed close to the working electrode in the cathodic compartment. Oxygen saturated 0.1 M NaOH solution ($\geq 99.99\%$ trace metals basis, semiconductor grade) served as the electrolyte. All experiments were performed with a sweep rate of 100 mV s⁻¹.

To convert the potential from the Ag/AgCl reference electrode to RHE the following equation was used: $E(RHE) = E(AgAgCl) + 0.197 + 0.0591 \times pH$, and the pH was determined based on full solvation of the 0.1 M NaOH, i.e. pH 13.

To calculate the TOF, the numbers of Co active sites used in the electrochemical reaction (n_{Co}) was first calculated as follows:

$$A_{Co} = \frac{A_{cell}}{3} = \frac{1}{3} \frac{\sqrt{3}}{2} s^2$$

$$n_{Co} = \frac{A_{crystal} \cdot \Theta}{A_{Co}} = \frac{\Theta \cdot \pi r^2}{\frac{\sqrt{3}}{6} s^2} = \frac{6\pi r^2}{\sqrt{3} s^2} \Theta$$

Where

A_{Co} : the average area between each Co atom;

A_{cell} : the area of the unit cell of the CoCN;

s : the unit cell distance, which is 11.5 Å, as measured with STM;

$A_{crystal}$: the area of the crystal exposed to electrolyte;

Θ : the coverage of the CoCN;

r : the radius of the crystal exposed to the electrolyte, 3 mm.

Here, we define the TOF as electrons transferred per cobalt atom per time. The TOF was calculated as follows:

$$TOF = \frac{e_{transferred}^-}{t \cdot n_{Co}} = \frac{i}{n_{Co}} = \frac{\sqrt{3} i s^2}{6\pi r^2 \Theta} \frac{e^-}{e}$$

Where

i : current for ORR, A or equivalently C s⁻¹;

e : The elementary charge

To get the number of molecules per time, divide the found TOF by 2 or 4 depending on the mix of H₂O₂ and H₂O formation.

The electrochemistry data is shown after the application of a Savitzky-Golay filter to smoothen experimental noise. For an example of unfiltered data see Figure S16.

ASSOCIATED CONTENT

Supporting Information.

The following files are available free of charge.

Supporting materials with figures and tables (PDF)

Figures S1-S14 containing additional characterization using STM and XPS, as well as reference measurements and more electrochemical data. Tables S1-S3 with details of XPS and DFT results (PDF)

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ABBREVIATIONS

XPS, X-ray Photoelectron Spectroscopy; STM, Scanning Tunneling Microscopy; TOF, Turn-over frequency; ORR, Oxygen Reduction Reaction; NP nanoparticle; RHE, Reversible Hydrogen Electrode

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TOC graphic

